

Phenotype data set
(public)

Phenotype ID	HIV status	Phenotype 1	Phenotype 2	...	Phenotype q
PID-1	HIV+	0.1	-2.7	...	90.3
PID-2	HIV-	0.5	8.6	...	63.5
⋮	⋮	⋮	⋮	⋮	⋮
PID- n	HIV-	-0.2	5.4	...	50.3

Phenotype-genotype
correlation data set

Phenotype 1 $\longleftrightarrow$ Variant 1
 Phenotype 2 $\longleftrightarrow$ Variant 2
 ...
 Phenotype q $\longleftrightarrow$ Variant q

1 Genotype data set
(stolen/hacked/queried)

Genotype ID	Variant 1	Variant 2	...	Variant q
GID-1	0	1	...	1
GID-2	2	1	...	0
⋮	⋮	⋮	⋮	⋮
GID- m	1	2	...	1

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Genotype
prediction

Phenotype ID	HIV status	Predicted variant genotypes			
		Variant 1	Variant 2	...	Variant q
PID-1	HIV+	1	0	...	2
PID-2	HIV-	2	2	...	1
⋮	⋮	⋮	⋮	⋮	⋮
PID- n	HIV-	0	1	...	1

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Genotype comparison
and matching

Predicted/matched
genotypes

Genotype ID	Phenotype ID	HIV status	Variant 1	Variant 2	...	Variant q
GID-1	PID-8	HIV+	0/0	1/1	...	1/1
GID-2	PID-3	HIV-	2/2	1/1	...	0/0
GID-3			1/0	1/0	...	0/2
GID-4	PID-1	HIV+	2/2	0/0	...	1/1
GID-5			0/1	1/1	...	2/1
⋮	⋮	⋮	⋮	⋮	⋮	⋮